Linear algebra section 3

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Quick warmup

1. If I have 7 equations and 9 variables, this system forms an $m \times n$ matrix before the augmentation. What are the values of m, n?

2. Can the above system have exactly one solution? (Hint: Think what happens to the rref and what lies after the augmentation line)

To get a sense of possible number of solutions of an $m \times n$ matrix, we look at its rank.

Definition: The rank of a matrix A is the number of leading 1's (pivots) in rref(A), denoted rank(A).

3. Consider the system of linear equations

$$\begin{cases} x + y = 2, \\ 2x + y = 3, \\ 3x + 2y = 5. \end{cases}$$

(a) Determine whether the system has a unique solution.

(b) If it does, find the solution.

Warmup:
4. If an $m \times n$ matrix has a unique solution and $n \neq m$, which of the following is true:
a. $m < n$
b. $m > n$